UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC662 mock paper (October 2026). Computed answers show working; theory answers are supported by lecture slides (not lecturer-marked) with a verbatim quote and slide citation.

2. Supported theory marking points quote the lecture decks; any unresolved slot would remain a visible placeholder.

3. Part A = 10 x (2 + 6); Part B = 2 x (5 + 5); total = 100 marks.

4. Computed: 2 slot(s). Supported theory: 22 slots. Placeholders: 0. Questions and answers are for local study use only.

5. Near-copy rejection used all 35 past papers; style widening is recorded per slot.

FOR MARKERS' USE ONLY

This marking scheme consists of 14 printed pages

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CSC662 - MARKING SCHEME (MOCK OCTOBER 2026)

Labels: computed / supported by lecture slides (not lecturer-marked) / placeholder. No official marking scheme exists for the real CSC662 papers; this is generated for the mock.

Entailment labels: supported = the quote directly supports the marking point (or the point passed the stem-overlap gate without an LLM check); partly supported = the quote supports part of the point or is ambiguous; not supported = the quote does not support the point. Only 'not supported' points triggered regeneration.

Part A: 10 x (2 + 6) = 80 marks. Part B: 2 x (5 + 5) = 20 marks. Total: 100 marks. Slots: 2 computed, 22 supported by lecture slides, 0 placeholder.

Methods: computed answers use verified solvers. Theory answers are supported by lecture slides (not lecturer-marked). Each marking point carries a verbatim quote, a deck/slide/chunk citation and a term-overlap score; low overlap is flagged for human review. Sources are restricted to the slot topic's allowed decks, and a slide is not reused across questions where an alternative exists.

PART A - QUESTION 1

A1a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) reasons why software is insecure according to the lecture.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Attackers can cause insecurity by altering resources that are essential to program execution, such as the instruction pointer or function call addresses.	"Software insecurity stems from attacker's ability to modify system resources criticial for program execution, e.g. instruction pointer, function call addresses, interrupt addresses, etc."	051 - Software Security, slide 30 051 - Software Security::s30::1	supported (stem gate)	0.67
2	A major cause of software insecurity is the failure to properly validate input supplied by users.	"One of the key sources for software insecurity is failed validation of user input."	051 - Software Security, slide 30 051 - Software Security::s30::1	supported (stem gate)	0.50

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 051 - Software Security, slide 30

A1b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Refer to your answer in (a). Explain six ways Windows security mechanisms manage users, objects and access, and why their complexity can itself become a security risk.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Windows offers elaborate built-in mechanisms covering user accounts, objects and access control.	"Windows systems contain complex security mechanisms for user account and object management as well as access control."	041 - Windows Security, slide 21 041 - Windows Security::s21::1	supported (stem gate)	0.64
2	Discretionary access control lists allow detailed control over diverse objects both in the OS and in applications.	"DACS enable fine-grained access control of heterogeneous entities in Windows itself as well as applications."	041 - Windows Security, slide 21 041 - Windows Security::s21::1	supported	0.30 LOW
3	Because the mechanisms are so complex, poor configuration or coding flaws can cause serious security breaches.	"The complexity of Windows security mechanisms is also its enemy: misconfiguration as well as implementation bugs may lead to severe security incidents."	041 - Windows Security, slide 21 041 - Windows Security::s21::1	supported	0.25 LOW

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
4	Active Directory, introduced with Windows 2000, is a directory service that organises users, groups, credentials, resources and policies.	"Active directory introduced in Windows 2000 is an LDAP-like directory service for organization of system resources:"	041 - Windows Security, slide 6 041 - Windows Security::s06::1	supported (stem gate)	0.58
5	Security attributes for a running subject are held in a token that maps the principal to the subject and is passed on to child processes.	"Security credentials for a subject are stored in a token."	041 - Windows Security, slide 11 041 - Windows Security::s11::1	partly supported	0.27 LOW
6	The registry stores security-relevant hives such as the SAM database, security logs and program paths.	"HKEY_LOCAL_MACHINE \ SAM: SAM database"	041 - Windows Security, slide 4 041 - Windows Security::s04::1	partly supported	0.18 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 041 - Windows Security, slide 11; 041 - Windows Security, slide 21; 041 - Windows Security, slide 4; 041 - Windows Security, slide 6

PART A - QUESTION 2

A2a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how a substitution cipher works and what serves as its key.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Encryption replaces each plaintext letter with a different letter to produce the ciphertext.	"In substitution ciphers, we replace the plain-text letters with other letters"	031 - Classical Cryptography, slide 4 031 - Classical Cryptography:: s04::1	supported	0.33 LOW
2	The substitution rule itself functions as the key of the cipher.	"The substitution rule is the key"	031 - Classical Cryptography, slide 4 031 - Classical Cryptography:: s04::1	supported	0.40 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 031 - Classical Cryptography, slide 4

A2b - 6 marks - computed - matches lecture slide 15 (standard tableau)

Question:

Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

Answer: SECURELOGINNOW

The 26 x 26 Vigenere tableau is printed in the question paper.

Working:

```
Standard 26 x 26 tableau, A=0; key repeated over the ciphertext.
Ciphertext: UMRBVVNWVPREQE
Key:        CIPHERCIPHERCI
Letter-by-letter: C - K mod 26 = P
  U (20) - C ( 2) = 18 -> S
  M (12) - I ( 8) =  4 -> E
  R (17) - P (15) =  2 -> C
  B ( 1) - H ( 7) = 20 -> U
  V (21) - E ( 4) = 17 -> R
  V (21) - R (17) =  4 -> E
  N (13) - C ( 2) = 11 -> L
  W (22) - I ( 8) = 14 -> O
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  V (21) - P (15) =  6 -> G
  P (15) - H ( 7) =  8 -> I
  R (17) - E ( 4) = 13 -> N
  E ( 4) - R (17) = 13 -> N
  Q (16) - C ( 2) = 14 -> O
  E ( 4) - I ( 8) = 22 -> W
Plaintext:  SECURELOGINNOW
```

Suggested marks: 1 mark for the key/table setup; 4 marks for the letter-by-letter working; 1 mark for the complete answer.

Confidence: computed - matches lecture slide 15 (standard tableau).

PART A - QUESTION 3

A3a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) formal security models that address integrity or conflicts of interest.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The Biba model, which adds integrity verification on top of access control.	"Biba model: additional integrity verification"	024 - Formal Security Model, slide 5 024 - Formal Security Model::s05::1	supported (stem gate)	0.50
2	The Chinese Wall model, which identifies conflicts of interest.	"Chinese Wall model: identification of conflicts of interest"	024 - Formal Security Model, slide 5 024 - Formal Security Model::s05::1	supported (stem gate)	1.00

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.

Citations: 024 - Formal Security Model, slide 5

A3b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six (6) points about the access control matrix, access control list and capability structures, covering how each is organised and its stated advantages or disadvantages.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The access control matrix is presented as a basic control structure.	"ACM is a basic control structure."	023 - Access Control and Authorization, slide 8 023 - Access Control and Authorization::s08::1	supported (stem gate)	0.60
2	The ACM is credited with clarity of definition and being easy to verify.	"Advantages: clarity of definition easy to verify"	023 - Access Control and Authorization, slide 8 023 - Access Control and Authorization::s08::1	supported (stem gate)	0.50
3	The ACM suffers from poor scalability and poor handling of changes.	"Disadvantages: poor scalability poor handling of changes"	023 - Access Control and Authorization, slide 8 023 - Access Control and Authorization::s08::1	supported (stem gate)	0.67

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
4	An ACL is an object-centred description of access rights, illustrated by entries such as bill.doc granting Bob read and write.	"ACL is an object-centered description of access rights: bill.doc: {Bob: read, write}"	023 - Access Control and Authorization, slide 10 023 - Access Control and Authorization::s10::1	supported (stem gate)	0.67
5	ACL advantages include easy access to object access rights and relatively easy management using abstractions.	"Advantages: easy access to object access rights relative easiness of management using abstractions"	023 - Access Control and Authorization, slide 10 023 - Access Control and Authorization::s10::1	supported (stem gate)	0.80
6	ACL disadvantages include poor overview of access rights per subject, difficulty of revocation and difficulty of sharing.	"Disadvantages: poor overview of access rights per subject difficulty of revocation difficulty of sharing"	023 - Access Control and Authorization, slide 10 023 - Access Control and Authorization::s10::1	supported (stem gate)	0.75

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 023 - Access Control and Authorization, slide 10; 023 - Access Control and Authorization, slide 8

PART A - QUESTION 4

A4a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List TWO (2) goals of access control and state what a reference monitor does.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	One goal of access control is granting access.	"Goals of access control: Granting access"	023 - Access Control and Authorization, slide 4 023 - Access Control and Authorization::s04::1	supported (stem gate)	1.00
2	A reference monitor is responsible for making authorization decisions.	"Reference monitor makes authorization decisions."	023 - Access Control and Authorization, slide 4 023 - Access Control and Authorization::s04::1	supported (stem gate)	0.83

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 023 - Access Control and Authorization, slide 4

Status: computed placeholder - no question generated for this type.

PART A - QUESTION 5

A5a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

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Explain two characteristics of Trojan horse malware that distinguish its behaviour on an infected host.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Trojans disguise themselves as legitimate or benign software so users install them willingly.	"They usually masquerade as helpful or harmless programs"	051 - Software Security, slide 12 051 - Software Security::s12::1	supported	0.00 LOW
2	Once running, a Trojan steals private data, deletes files, records keystrokes, or opens a backdoor for additional malware.	"begin to steal private data, delete files, log keystrokes or provide an opening for further malware to get installed"	051 - Software Security, slide 12 051 - Software Security::s12::1	supported (stem gate)	0.57

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 12

A5b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six techniques by which malware infects a host or propagates itself, and the effect of each technique.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	A boot sector infector saves the original master boot record elsewhere before overwriting it with an infected version, so the system then boots using the infected MBR.	"Save the original MBR in a safe location"	051 - Software Security, slide 8 051 - Software Security::s08::1	partly supported	0.20 LOW
2	A COM-file infector appends the virus body to the program and replaces the program's entry point with a jump into the virus body.	"Replace a program entry point with a jump to a virus body"	051 - Software Security, slide 9 051 - Software Security::s09::1	supported (stem gate)	0.58
3	A COM-file infector restores the original entry point and transfers control back to it once the virus code has finished executing.	"Virus code restores the original entry point and jumps to it after its own execution"	051 - Software Security, slide 9 051 - Software Security::s09::1	supported	0.44 LOW
4	An EXE-file infector overwrites the program header so that the entry point is switched to the virus.	"Over write a program header to switch the entry point to a virus"	051 - Software Security, slide 10 051 - Software Security::s10::1	supported (stem gate)	0.60
5	An EXE-file infector lets the infected program jump to its original entry point during execution, so the host still appears to run normally.	"Jump to the original entry point during execution"	051 - Software Security, slide 10 051 - Software Security::s10::1	partly supported	0.35 LOW
6	A rootkit is typically installed after a vulnerability has been found and detected, so that the infected computer stays under the malware author's control.	"They are usually installed after a vulnerability has been found and detected so that the infected computer continues to be under the control of the malware author."	051 - Software Security, slide 13 051 - Software Security::s13::1	supported (stem gate)	0.80

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 10; 051 - Software Security, slide 13; 051 - Software Security, slide 8; 051 - Software Security, slide 9

PART A - QUESTION 6

A6a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State the two stages of the biometric authentication process.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The identification stage, which involves acquisition, creation of master characteristics and storage of master characteristics.	"Identification Stage Acquisition Creation of Master characteristics Storage of Master characteristics"	072 - Biometrics and Authentication, slide 5 072 - Biometrics and Authentication:: s05::1	supported (stem gate)	0.86
2	The authentication stage, which involves acquisition(s), comparison and decision.	"Authentication Stage Acquisition(s) Comparison Decision"	072 - Biometrics and Authentication, slide 5 072 - Biometrics and Authentication:: s05::1	supported (stem gate)	0.83

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 072 - Biometrics and Authentication, slide 5

A6b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six facts about biometric mechanisms and technologies as presented in the course, covering the definition, the categories, and specific biometric technologies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	A biometric is defined as a physiological or behavioural characteristic of a human being that can distinguish one person from another and can theoretically be used for identification or verification of identity.	"A biometric is a physiological or behavioural characteristic of a human being that can distinguish one person from another and that theoretically can be used for identification or verification of identity."	072 - Biometrics and Authentication, slide 2 072 - Biometrics and Authentication:: s02::1	supported (stem gate)	0.92
2	Biometric applications available today are divided into two sectors: psychological and behavioural.	"Biometric applications available today are categorized into 2 sectors"	072 - Biometrics and Authentication, slide 4 072 - Biometrics and Authentication:: s04::1	supported (stem gate)	0.67
3	The psychological sector includes iris, fingerprints, hand, retinal and face recognition.	"Psychological: Iris, Fingerprints, Hand, Retinal and Face recognition"	072 - Biometrics and Authentication, slide 4 072 - Biometrics and Authentication:: s04::1	supported (stem gate)	0.78
4	The behavioural sector includes voice, typing pattern and signature.	"Behavioural: Voice, Typing pattern, Signature"	072 - Biometrics and Authentication, slide 4 072 - Biometrics and Authentication:: s04::1	supported (stem gate)	0.67
5	Fingerprint processing divides the print into loops, whorls and arch and calculates minutiae points (ridge endings).	"Divides print into loops, whorls and arch"	072 - Biometrics and Authentication, slide 6 072 - Biometrics and Authentication:: s06::1	partly supported	0.46 LOW
6	Iris scanning scans the unique pattern of the iris, which is coloured and visible from far, and requires no touch, thereby overcoming retinal scanner issues.	"Scans unique pattern of iris"	072 - Biometrics and Authentication, slide 10 072 - Biometrics and Authentication:: s10::1	partly supported	0.29 LOW

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Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 072 - Biometrics and Authentication, slide 10; 072 - Biometrics and Authentication, slide 2; 072 - Biometrics and Authentication, slide 4; 072 - Biometrics and Authentication, slide 6

PART A - QUESTION 7

A7a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain where NIDS sensors should be placed and why such placement matters.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Sensors should be positioned deliberately so that the most valuable assets are defended.	"Place IDS sensors strategically to defend most valuable assets"	062 - Intrusion Detection System, slide 7 062 - Intrusion Detection System::s07::1	supported (stem gate)	0.62
2	A typical sensor location is just inside the firewall.	"Just inside the firewall"	062 - Intrusion Detection System, slide 7 062 - Intrusion Detection System::s07::1	supported (stem gate)	0.50

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 062 - Intrusion Detection System, slide 7

A7b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six preventive network security mechanisms and, for each, one function it performs in protecting an organization's network.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Firewall: it controls incoming and outgoing network traffic according to predetermined security policies.	"Control incoming and outgoing traffic on network with predetermined security policies."	061 - Network Security, slide 3 061 - Network Security::s03::1	supported (stem gate)	0.80
2	Proxy server: it improves data security and network performance by acting as a firewall and web filter, sharing connections and caching data.	"Proxy servers act as a firewall and web filter, provide shared network connections, and cache data to speed up common requests."	061 - Network Security, slide 3 061 - Network Security::s03::1	supported (stem gate)	0.79
3	Network segmentation: it creates boundaries between network segments whose computing assets share a common function, risk or role.	"Boundaries between network segments where computing assets within the segment have a common function, risk or role within an organization."	061 - Network Security, slide 3 061 - Network Security::s03::1	supported (stem gate)	0.77
4	Remote access VPN: it gives individual hosts or clients remote and secure access to the organization's network.	"Provides remote and secure access to an organization network to individual hosts or clients"	061 - Network Security, slide 4 061 - Network Security::s04::1	supported (stem gate)	0.89
5	Zero Trust Network Access (ZTNA): a user is granted only the access and permission required to fulfil the role.	"A user should only has access and permission that required to fulfill the role"	061 - Network Security, slide 4 061 - Network Security::s04::1	supported (stem gate)	0.58
6	Intrusion Detection/Prevention Systems (IDS/IPS): they detect or prevent attacks such as brute force, Denial-of-Service and exploits of network vulnerabilities.	"To detect or prevent the network security attacks such as brute force attacks, Denial-of-Service attacks and exploits of network vulnerabilities."	061 - Network Security, slide 4 061 - Network Security::s04::1	supported (stem gate)	0.69

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited

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lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 061 - Network Security, slide 3; 061 - Network Security, slide 4

PART A - QUESTION 8

A8a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain two password security controls used to protect password files and accounts.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Access control should be applied to password files to protect them.	"Access control applied to password files"	022 - Identification and Authentication, slide 12 022 - Identification and Authentication:: s12::1	supported (stem gate)	0.50
2	Accounts should be de-activated after multiple failed login attempts.	"De-activation of accounts after multiple failed logins"	022 - Identification and Authentication, slide 12 022 - Identification and Authentication:: s12::1	supported (stem gate)	0.75

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 022 - Identification and Authentication, slide 12

A8b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss three design criteria of Kerberos and three motivations for adopting single sign-on in distributed environments.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Kerberos must prevent eavesdropping by ensuring that passwords are never transmitted in clear text.	"Security against eavesdropping: no password transmission in clear text."	022 - Identification and Authentication, slide 18 022 - Identification and Authentication:: s18::1	supported	0.40 LOW
2	Kerberos must guarantee reliability by requiring prior authentication for every use of a service.	"Reliability: every use of a service requires prior authentication."	022 - Identification and Authentication, slide 18 022 - Identification and Authentication:: s18::1	supported (stem gate)	0.67
3	Kerberos must be transparent so that the user is unaware of any authentication beyond the initial login.	"Transparency: user is not aware of any authentication beyond an initial login."	022 - Identification and Authentication, slide 18 022 - Identification and Authentication:: s18::1	supported (stem gate)	0.70
4	Kerberos must be scalable to support a large number of servers and clients.	"Scalability: support for a large number of servers and clients."	022 - Identification and Authentication, slide 18 022 - Identification and Authentication:: s18::1	supported (stem gate)	0.75

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
5	Single sign-on reduces the burden on users to memorize passwords.	"Reduce the need to memorize passwords"	022 - Identification and Authentication, slide 16 022 - Identification and Authentication:: s16::1	supported	0.43 LOW
6	Single sign-on reduces the time users spend typing passwords.	"Reduce the time spent on typing passwords"	022 - Identification and Authentication, slide 16 022 - Identification and Authentication:: s16::1	supported	0.43 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 022 - Identification and Authentication, slide 16; 022 - Identification and Authentication, slide 18

PART A - QUESTION 9

A9a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how a chain of trust is established in trusted computing using the TPM.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Each stage of the chain measures and validates the following stage.	"Each stage measures and validates the next one"	073 - Trusted Computing, slide 8 073 - Trusted Computing::s08::1	supported (stem gate)	0.60
2	The measurements produced are stored in Platform Configuration Registers on the TPM.	"Measurements go into Platform Configuration Registers (PCRs) on the TPM"	073 - Trusted Computing, slide 8 073 - Trusted Computing::s08::1	supported (stem gate)	0.71

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 073 - Trusted Computing, slide 8

A9b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain six components and functions of the NGSCB architecture, focusing on the Nexus and the Nexus Computing Agent.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The Nexus is a security kernel that is authenticated during boot.	"Security kernel, authenticated on boot"	073 - Trusted Computing, slide 14 073 - Trusted Computing::s14::1	supported (stem gate)	0.67
2	The Nexus provides an application interface to the TPM.	"Application interface to TPM"	073 - Trusted Computing, slide 14 073 - Trusted Computing::s14::1	supported (stem gate)	0.60
3	The Nexus Computing Agent is a trusted software component.	"Trusted software component"	073 - Trusted Computing, slide 15 073 - Trusted Computing::s15::1	supported	0.40 LOW

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
4	The NCA runs in a trusted mode that communicates with the Nexus.	"Runs in trusted mode that communicates with Nexus"	073 - Trusted Computing, slide 15 073 - Trusted Computing::s15::1	supported (stem gate)	0.83
5	The Nexus is a special kernel forming the core of the trusted operating environment.	"Special kernel (core of the trusted operating)"	073 - Trusted Computing, slide 13 073 - Trusted Computing::s13::1	supported (stem gate)	0.62
6	The Nexus isolates normal-mode processes from trusted-mode processes differently in memory.	"Isolate the process of normal mode and trusted mode differently in memory"	073 - Trusted Computing, slide 13 073 - Trusted Computing::s13::1	supported (stem gate)	0.86

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 073 - Trusted Computing, slide 13; 073 - Trusted Computing, slide 14; 073 - Trusted Computing, slide 15

PART A - QUESTION 10

A10a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List two impacts of threats on confidentiality, integrity or availability.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Disclosure is a threat class involving unauthorized access to information, which harms confidentiality.	"Disclosure - unauthorized access to information"	011 - Introduction to Computer Security, slide 10 011 - Introduction to Computer Security::s10::1	supported	0.44 LOW
2	Deception involves acceptance of false data, which harms integrity.	"Deception - acceptance of false data"	011 - Introduction to Computer Security, slide 10 011 - Introduction to Computer Security::s10::1	supported (stem gate)	0.57

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 011 - Introduction to Computer Security, slide 10

A10b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how the Samy worm used cross-site scripting to infect MySpace user profiles, and what made the attack so widespread.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The worm injected JavaScript into a user profile by writing to document.body.innerHTML.	"Inject Javascript into a user profile using document.body.innerHTML"	052 - Web Security, slide 15 052 - Web Security::s15::1	supported (stem gate)	0.70
2	It retrieved the list of the user's heroes by issuing a GET request to a user profile.	"Get the list of user's heros by using a GET on a user profile"	052 - Web Security, slide 15 052 - Web Security::s15::1	supported	0.43 LOW
3	It added Samy as a friend by performing an XML-HTTP request against the addFriends page.	"Add Samy as a field by performing XML-HTTP on the addFriends page"	052 - Web Security, slide 15 052 - Web Security::s15::1	supported (stem gate)	0.70

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
4	Any visitor to an infected profile had their own profile and friends list infected, enabling self-propagation.	"Any user who visits an infected user's page will gets profile infected with Javascript and his friends list infected with Samy"	052 - Web Security, slide 15 052 - Web Security::s15::1	supported	0.30 LOW
5	The worm reached a very large scale, infecting roughly one million MySpace profiles.	"Eventually about 1,000,000 MySpace profiles were infected."	052 - Web Security, slide 15 052 - Web Security::s15::1	supported	0.20 LOW
6	The underlying weakness is that any web application returning user input without filtering is a potential XSS vulnerability.	"Any web application that returns user input without filtering is a potential XSS vulnerability!"	052 - Web Security, slide 13 052 - Web Security::s13::1	supported (stem gate)	0.77

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 052 - Web Security, slide 13; 052 - Web Security, slide 15

PART B - QUESTION 1

B1a - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how an intrusion detection system contributes to detecting and defending against attacks, and outline the limitations of misuse-based detection.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	An IDS detects malicious activity occurring in computer systems.	"Detects malicious activity in computer systems"	062 - Intrusion Detection System, slide 2 062 - Intrusion Detection System:: s02::1	supported (stem gate)	0.71
2	An IDS can identify attacks that are currently in progress and stop them.	"Identifies and stops attacks in progress"	062 - Intrusion Detection System, slide 2 062 - Intrusion Detection System:: s02::1	supported	0.43 LOW
3	An IDS supports forensic analysis after an attack has finished.	"Conducts forensic analysis once attack is over"	062 - Intrusion Detection System, slide 2 062 - Intrusion Detection System:: s02::1	supported	0.38 LOW
4	Misuse detection represents attacks as patterns or signatures so that variations of the same attack can still be recognised.	"there are ways to represent attacks in the form of a pattern or a signature so that even variations of the same attack can be detected"	062 - Intrusion Detection System, slide 17 062 - Intrusion Detection System:: s17::1	supported (stem gate)	0.73
5	Misuse detection is of little use against unknown attack methods, which is a key limitation.	"They can detect many or all known attack patterns, but they are of little use for unknown attack methods."	062 - Intrusion Detection System, slide 17 062 - Intrusion Detection System:: s17::1	supported (stem gate)	0.62

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 062 - Intrusion Detection System, slide 17; 062 - Intrusion Detection System, slide 2

B1b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the distinction between a vulnerability, a threat and an attack, and describe how organisations counter threats arising from forces of nature and from malicious web input.

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Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	A vulnerability is a weakness in a system that can be exploited to damage its assets.	"Vulnerabilities: Weakness of a system that can be exploited to damage assets"	011 - Introduction to Computer Security, slide 9 011 - Introduction to Computer Security::s09::1	supported (stem gate)	1.00
2	A threat is only a potential violation of security, not yet an actual one.	"Threats: A threat is a "potential" violation of security"	011 - Introduction to Computer Security, slide 9 011 - Introduction to Computer Security::s09::1	supported (stem gate)	0.50
3	An attack is the actual violation of security that has occurred.	"Attacks: The actual violation of security"	011 - Introduction to Computer Security, slide 9 011 - Introduction to Computer Security::s09::1	supported (stem gate)	0.67
4	Forces of nature are counted among the most dangerous threats to information systems.	"Forces of nature are among the most dangerous threats"	011 - Introduction to Computer Security, slide 13 011 - Introduction to Computer Security::s13::1	supported (stem gate)	0.67
5	Organisations must put controls in place to limit the damage from such threats and prepare contingency plans so operations can continue.	"Organizations must implement controls to limit damage and prepare contingency plans for continued operations"	011 - Introduction to Computer Security, slide 13 011 - Introduction to Computer Security::s13::1	supported (stem gate)	0.60

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 011 - Introduction to Computer Security, slide 13; 011 - Introduction to Computer Security, slide 9

PART B - QUESTION 2

B2a - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

State the meaning of morals, ethics and laws, and state two limitations of relying on a professional code of ethics.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Morals are described as traditions of belief concerning right and wrong conduct.	"Morals are traditions of belief about right and wrong conduct"	071 - Ethics and Computer Security, slide 5 071 - Ethics and Computer Security::s05::1	supported (stem gate)	0.86
2	Ethics is characterised as a suite of guiding beliefs, standards or ideals that pervades an individual or a group or community.	"Ethics is a suite of guiding beliefs, standards, or ideals that pervades an individual or a group or community of people"	071 - Ethics and Computer Security, slide 5 071 - Ethics and Computer Security::s05::1	supported (stem gate)	0.91
3	Laws are defined as formal rules of conduct imposed by a sovereign authority such as a government on its subjects or citizens.	"Laws are formal rules of conduct that a sovereign authority, such as a government, imposes on its subjects or citizens"	071 - Ethics and Computer Security, slide 5 071 - Ethics and Computer Security::s05::1	supported (stem gate)	0.92

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
4	A code of ethics may fail to reflect the ethical views of every member of the group.	"May not reflect the ethical of every member of the group."	071 - Ethics and Computer Security, slide 9 071 - Ethics and Computer Security::s09::1	supported (stem gate)	0.67
5	A code of ethics is often not enforceable because it is a code of conduct rather than a legal document.	"Many times it isn't enforceable. It's a code of conduct, not a legal document."	071 - Ethics and Computer Security, slide 9 071 - Ethics and Computer Security::s09::1	supported (stem gate)	0.55

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 5; 071 - Ethics and Computer Security, slide 9

B2b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss what a professional code of ethics and conduct for computing staff should contain and how it should guide their behaviour, with reference to recognised codes and the ethical issues they address.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	State that a code of ethics is a documented set of acceptable behaviours for a professional or social group.	"Documented set of acceptable behaviors for professional or social groups."	071 - Ethics and Computer Security, slide 6 071 - Ethics and Computer Security::s06::1	supported (stem gate)	0.67
2	Note that recognised bodies such as the ACM have their own code of ethics and professional conduct.	"Association of Computing Machinery ACM (founded 1947) has a code of ethics and professional conduct."	071 - Ethics and Computer Security, slide 7 071 - Ethics and Computer Security::s07::1	supported	0.46 LOW
3	Give an example of an ACM ethical principle, such as contributing to society and human well-being.	"Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing."	071 - Ethics and Computer Security, slide 11 071 - Ethics and Computer Security::s11::1	supported	0.44 LOW
4	Give an example of ACM professional conduct, such as maintaining high standards of competence, conduct and ethical practice.	"Maintain high standards of professional competence, conduct, and ethical practice."	071 - Ethics and Computer Security, slide 12 071 - Ethics and Computer Security::s12::1	supported (stem gate)	0.73
5	Explain that the code should address privacy by protecting a person's personal information.	"Protecting one's personal information"	071 - Ethics and Computer Security, slide 13 071 - Ethics and Computer Security::s13::1	supported	0.43 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 11; 071 - Ethics and Computer Security, slide 12; 071 - Ethics and Computer Security, slide 13; 071 - Ethics and Computer Security, slide 6; 071 - Ethics and Computer Security, slide 7

End of marking scheme.

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